

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12415360
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Iida; Shotaro et al.

Controller, control method, and image recording device

Abstract

There is provided a controller for an image recording device. The image recording device includes: a carriage capable of moving in a first direction; a head mounted on the carriage and configured to discharge a liquid from a nozzle; and a wiper capable of being impregnated with a cleaning liquid, a nozzle surface of the head including a first region not having the nozzle and a second region having the nozzle. The controller is configured to execute: a pressing-against processing of abutting the wiper on the first region in a state in which a speed of a relative movement in the first direction between the carriage and the wiper is a first speed; and a moving processing of moving the carriage and the wiper relative to each other in the first direction at a second speed faster than the first speed, while abutting the wiper on the second region.

Inventors:	Iida; Shotaro (Nagoya, JP), Taira; Hiroshi (Ichinomiya, JP)
Applicant:	BROTHER KOGYO KABUSHIKI KAISHA (Nagoya, JP)
Family ID:	1000008819238
Assignee:	BROTHER KOGYO KABUSHIKI KAISHA (Nagoya, JP)
Appl. No.:	18/055871
Filed:	November 16, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230150266 A1	May. 18, 2023

Foreign Application Priority Data

JP	2021-186669	Nov. 16, 2021
----	-------------	---------------

Publication Classification

Int. Cl.: B41J2/165 (20060101); B41J2/045 (20060101); B41J2/14 (20060101)

U.S. Cl.:

CPC B41J2/16552 (20130101); B41J2/04503 (20130101); B41J2002/14322 (20130101)

Field of Classification Search

CPC: B41J (2/16552); B41J (2/04503); B41J (2002/14322); B41J (2002/16558); B41J (2/16517); B41J (2/16541); B41J (2/16538)

USPC: 347/22

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6460967	12/2001	Makita et al.	N/A	N/A
2002/0171705	12/2001	Rhoads	347/33	B41J 2/16538
2005/0231548	12/2004	Tsukada	347/33	B41J 2/16535

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
H11-334095	12/1998	JP	N/A

Primary Examiner: Uhlenhake; Jason S

Assistant Examiner: Shenderov; Alexander D

Attorney, Agent or Firm: Scully, Scott, Murphy & Presser, P.C.

Background/Summary

REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority from Japanese Patent Application No. 2021-186669 filed on Nov. 16, 2021. The entire content of the priority application is incorporated herein by reference.

BACKGROUND ART

(2) There is known an image recording device comprising a head and a carriage mounted with the head, that discharges ink from nozzles of the head when the carriage moves in a predetermined direction. In the image recording device, a wipe processing in which a nozzle surface of the head is wiped by a wiper is executed to remove an unnecessary object (the likes of ink or a contaminant) that has adhered to the nozzle surface.

(3) As the wipe processing, there is known a method by which a cleaning liquid is caused to adhere to the nozzle surface by the wiper that has been impregnated with the cleaning liquid, and the cleaning liquid or unnecessary object that have adhered to the nozzle surface are wiped away. Sometimes, in this method, if an amount of the cleaning liquid provided to the nozzle surface is too large, then when the wiper separates from the nozzle surface, the cleaning liquid or unnecessary object that have been wiped away by the wiper will scatter. In this context, there is known a liquid discharge device in which a squeezing roller is pressed against a wiping member having liquid

impregnatable properties so as to squeeze the wiping member.

DESCRIPTION

(4) In the liquid discharge device described in Japanese Patent Publication Laid-Open No. H11-334095, a plurality of dedicated members is required to press the squeezing roller against the wiping member, hence configuration of the device becomes complicated.

(5) The present disclosure, which was made in view of the above-described circumstances, has an object of providing a means by which wipe processing employing a suitable amount of a cleaning liquid can be executed by a simple configuration.

(6) (1) A controller according to the present disclosure is a controller for an image recording device, the image recording device including: a carriage capable of moving in a first direction; a head which is mounted on the carriage and which is configured to discharge a liquid from a nozzle; and a wiper capable of being impregnated with a cleaning liquid, a nozzle surface of the head including a first region not having the nozzle and a second region having the nozzle, the controller being configured to execute: a pressing-against processing of abutting the wiper on the first region in a state in which a speed of a relative movement in the first direction between the carriage and the wiper is a first speed; and a moving processing of moving the carriage and the wiper relative to each other in the first direction at a second speed faster than the first speed, while abutting the wiper on the second region.

(7) The above-described controller results in that, by a pressing-against processing being executed, a wiper impregnated with a cleaning liquid is squeezed, and the cleaning liquid impregnating the wiper can be reduced to a suitable amount. The pressing-against processing can be executed without a dedicated member being employed. Hence, by executing a moving processing after the pressing-against processing, a wipe processing employing a suitable amount of the cleaning liquid can be executed by a simple configuration.

(8) (2) The first speed may be zero.

(9) (3) The first speed may be greater than zero.

(10) (4) The image recording device may further include an additional wiper not capable of being impregnated with the cleaning liquid; and in the moving processing, the carriage and the additional wiper are moved relative to each other in the first direction at the second speed, in a state in which the additional wiper abuts on the second region.

(11) (5) The pressing-against processing may be executed in a state in which the additional wiper abuts on a side surface of the head.

(12) (6) In the pressing-against processing, the carriage and the wiper may be moved relative to each other in a moving direction; and then, in the moving processing, the carriage and the wiper may be moved relative to each other in the moving direction, without being moved relative to each other in a direction opposite to the moving direction.

(13) (7) The controller may be configured to: after the moving processing, move the carriage and the wiper relative to each other, in a state in which the wiper does not abut on the head, until the wiper is located relative to the head at a starting position or at a position further from an ending position as compared with the starting position, the starting position being a position of the wiper relative to the head when the moving processing is started, the ending position being a position of the wiper relative to the head when the moving processing is ended; and execute the moving processing again.

(14) (8) The controller may be configured to execute the pressing-against processing for one second or more.

(15) (9) A control method according to the present disclosure is a control method for an image recording device, the image recording device including: a carriage capable of moving in a first direction; a head which is mounted on the carriage and which is configured to discharges a liquid from a nozzle; and a wiper capable of being impregnated with a cleaning liquid, a nozzle surface of the head including a first region not having the nozzle and a second region having the nozzle, the

control method comprising: executing a pressing-against processing of abutting the wiper on the first region in a state in which a speed of a relative movement in the first direction between the carriage and the wiper is a first speed; and executing a moving processing of moving the carriage and the wiper relative to each other in the first direction at a second speed faster than the first speed, while abutting the wiper on the second region.

(16) (10) An image recording device according to the present disclosure includes: a carriage capable of moving in a first direction; a head which is mounted on the carriage and which is configured to discharge a liquid from a nozzle; a wiper capable of being impregnated with a cleaning liquid; and a controller, wherein a nozzle surface of the head includes a first region not having a nozzle and a second region having a nozzle, and the controller is configured to execute: a pressing-against processing of abutting the wiper on the first region in a state in which a speed of a relative movement in the first direction between the carriage and the wiper is a first speed; and a moving processing of moving the carriage and the wiper relative to each other in the first direction at a second speed faster than the first speed, while abutting the wiper on the second region.

(17) (11) A controller according to the present disclosure is a controller for an image recording device, the image recording device including: a carriage capable of moving in a first direction; a head which is mounted on the carriage and which is configured to discharge a liquid from a nozzle; and a wiper capable of being impregnated with a cleaning liquid, a nozzle surface of the head including a first region not having the nozzle and a second region having the nozzle, the controller being configured to execute: a pressing-against processing of abutting the wiper on the first region to deform the wiper such that at least part of the cleaning liquid absorbed in the wiper is separated from the wiper; and a wiping processing of wiping the second region with the wiper after the pressing-against processing.

(18) (12) The wiping processing may include wiping the second region with the wiper while moving the head and the wiper relative to each other in the first direction, the wiper may be a plate member, a thickness direction of the wiper being the first direction, and the wiper may be configured to bend by a force in the first direction applied by the head in the pressing-against processing.

(19) (13) The pressing-against processing may include abutting the wiper on the first region in a state in which a speed of a relative movement in the first direction between the carriage and the wiper is a first speed; and the wiping processing may include moving the carriage and the wiper relative to each other in the first direction at a second speed faster than the first speed.

(20) (14) The first speed may be zero.

Description

(1) The present disclosure results in that a wipe processing employing a suitable amount of a cleaning liquid can be executed by a simple configuration.

(2) FIG. 1 is a schematic view depicting internal configuration of a printer 10.

(3) FIG. 2 is a diagram depicting a moving range of a carriage 41.

(4) FIG. 3 is a block diagram depicting configuration of a controller 60 and elements connected to the controller 60.

(5) FIG. 4 is a perspective view of a wiper unit 50.

(6) FIGS. 5A, 5B, and 5C are schematic views depicting states of the wiper unit 50.

(7) FIG. 5A depicts an upward facing state, FIG. 5B depicts an intermediate state, and FIG. 5C depicts a downward facing state.

(8) FIG. 6 is a flowchart depicting operation of the controller 60.

(9) FIG. 7 is a flowchart of a wipe processing by the controller 60.

(10) FIGS. 8A, 8B and 8C are schematic views depicting states of two wipers of the wiper unit 50.

FIG. 8A depicts a state before a pressing-against processing, FIG. 8B depicts a state during the pressing-against processing, and FIG. 8C depicts a state during a moving processing.

(11) FIG. 9 is a flowchart of a wipe processing by a controller.

(12) FIG. 10 is a perspective view of a wiper unit 59.

(13) A printer 10 (one example of an image recording device) and controller 60 (one example of a controller) according to an embodiment of the present disclosure will be described below. Note that the embodiment described below is merely one example of the present disclosure, and it goes without saying that the embodiment of the present disclosure can be appropriately altered in a range not altering the gist of the present invention. In the description below, an up-down direction 7 is defined with reference to a state of the printer 10 having been useably installed (the state of FIG. 1), a front-rear direction 8 is defined assuming a surface provided with a discharge port 13 to be a front surface, and a left-right direction 9 is defined looking at the printer 10 from the front. The up-down direction 7, the front-rear direction 8, and the left-right direction 9 are orthogonal to each other.

Overall Configuration of Printer 10

(14) The printer 10 depicted in FIG. 1 is an image recording device that records an image on a sheet S by an ink jet recording system. The sheet S is a long paper sheet that has been wound into a roll shape. In order for the sheet S to be mounted in the printer 10, a winding center of the sheet S has a through-hole formed therein. A recording objective medium may be the likes of a seal paper, fan-fold paper, cut paper, or a fabric.

(15) The printer 10 includes a housing 11 of substantially rectangular parallelepiped shape. The housing 11 has a size enabling it to be mounted on a tabletop, on a floor, or in a rack, and so on. Located in a front wall 12 of the housing 11 is the discharge port 13 which is slit-like and extends in the left-right direction 9. The sheet S that has been recorded with an image by the printer 10 is discharged from the discharge port 13. The discharged sheet S is wound by a winding device (not depicted) attached to the printer 10, for example.

(16) As depicted in FIG. 1, the printer 10 includes the following within the housing 11, namely, a holder 21, a tensioner 22, a conveying roller pair 23, a discharging roller pair 24, a platen 25, four tanks 26A-26D, a carriage 41, and a head 42. The head 42 is mounted on the carriage 41. As depicted in FIG. 2, the printer 10 further includes the following within the housing 11, namely, two guide rails 37, 38, a cap 39, and a wiper unit 50. As depicted in FIG. 3, the printer 10 further includes the following within the housing 11, namely, the controller 60, a motor for holder drive 71, a motor for conveyance 72, a motor for carriage drive 73, a motor for cap drive 74, and a motor for wiper drive 75. The printer 10 may further include various kinds of sensors, and so on, besides the above-mentioned elements.

Tanks 26A-26D

(17) The tanks 26A-26D respectively store yellow, magenta, cyan, and black inks. The inks, which are so-called latex inks, contain a pigment, resin fine particles, and an additive. The ink has a viscosity suitable for evenly dispersing the pigment and the resin fine particles. The pigment represents the color of the ink. The resin fine particles, which are for causing the pigment to adhere to the sheet S, are of a synthetic resin whose glass transition temperature is exceeded by heating of a heater (not depicted), for example.

(18) Note that the printer 10 may include at least one tank. Moreover, the tank may store a liquid other than ink. The liquid stored in the tank is a preprocessing liquid, for example. The preprocessing liquid may contain a cationic polymer, a polyvalent metal salt (for example, a magnesium salt), or the like. The preprocessing liquid has a function of preventing running or bleeding-through of ink, due to it causing components in the ink to aggregate or precipitate. In some cases, the preprocessing liquid has a function of improving coloring or quick-drying properties of the ink.

Conveying Mechanism of Sheet S

(19) Located inside the housing **11** are a pair of side frames (not depicted) that extend in the up-down direction **7** and the front-rear direction **8**. The holder **21** has a rotating shaft **31** that supports the sheet **S**. The rotating shaft **31** extends in the left-right direction **9** and has its two ends fixed to the side frames. Motive power of the motor for holder drive **71** (see, FIG. **3**) is transmitted to the rotating shaft **31**. This motive power causes the holder **21** to rotate in a circumferential direction of the rotating shaft **31**. In FIG. **1**, a rotating direction of the holder **21** is counter-clockwise. Rotation of the holder **21** causes a roll body supported by the holder **21** to rotate too. The sheet **S** is led out upwardly from a rear end of the roll body and guided to the tensioner **22** as a result of the conveying roller pair **23** and discharging roller pair **24** rotating

(20) The tensioner **22**, the conveying roller pair **23**, and the discharging roller pair **24** each extend in the left-right direction **9** between the side frames and are installed in a manner enabling them to rotate in a circumferential direction of their rotational axes parallel to the left-right direction **9**. The tensioner **22** is applied with a biasing force in a rearward direction by a biasing member such as a spring. The tensioner **22** abuts on the sheet **S** led out from the roll body and thereby guides the sheet **S** in such a manner that the sheet **S** curves frontwards.

(21) The conveying roller pair **23** include a drive roller **32** and a pinch roller **33**, and are located forward of the tensioner **22**. The discharging roller pair **24** include a drive roller **34** and a pinch roller **35**, and are located further forward than the conveying roller pair **23**. Lower end positions of the drive rollers **32**, **34** substantially coincide with an upper end position of the tensioner **22** in the up-down direction **7**. The pinch roller **33** abuts from below on the drive roller **32**. The pinch roller **35** abuts from below on the drive roller **34**.

(22) Motive power of the motor for conveyance **72** (see, FIG. **3**) is transmitted to the drive rollers **32**, **34**. This motive power causes the drive rollers **32**, **34** to rotate. As a result, the drive rollers **32**, **34** nip the sheet **S** between themselves and the pinch rollers **33**, **35**, and, while doing so, convey the sheet **S** in a conveying direction **6**. In the present embodiment, the conveying direction **6** is a frontward direction.

Platen **25**

(23) The platen **25** is attached to the side frames at a position between the conveying roller pair **23** and discharging roller pair **24** in the front-rear direction **8**. The platen **25** extends in the left-right direction **9** between the side frames, and includes a supporting surface **36** of the sheet **S**, that extends in the front-rear direction **8** and the left-right direction **9**. The supporting surface **36** is an upper end surface of the platen **25**. A position in the up-down direction **7** of the supporting surface **36** substantially coincides with the upper end position of the tensioner **22**. The platen **25** may be a sucking platen that sucks the sheet **S** onto the supporting surface **36**.

Carriage **41** and Head **42**

(24) As depicted in FIG. **2**, the guide rails **37**, **38** extend in the left-right direction **9** parallelly to each other. Positions in the up-down direction **7** of the guide rails **37**, **38** are the same. The guide rail **38** is located rearward of the guide rail **37** in the front-rear direction **8**. Both ends of the guide rails **37**, **38** are fixed to the side frames. The carriage **41** is supported by the guide rails **37**, **38**. Motive power of the motor for carriage drive **73** (see, FIG. **3**) is transmitted to a carriage drive mechanism (not depicted). The carriage **41** moves in the left-right direction **9** due to action of the carriage drive mechanism, in a state of being supported by the guide rails **37**, **38**. The left-right direction **9** is an example of a first direction.

(25) As depicted in FIG. **1**, the head **42** is mounted on the carriage **41**. A lower surface of the head **42** is referred to as a nozzle surface **43**. A plurality of nozzles **44** each configured to discharging ink is formed in the nozzle surface **43**. The tanks **26A-26D** and the head **42** are connected via ink channels (not depicted). Inks stored in the tanks **26A-26D** are supplied to the head **42** via the ink channels. While the carriage **41** is moving in the left-right direction **9**, inks that have been supplied to the head **42** are discharged from the nozzles **44**. As a result, image recording is performed on the sheet **S**.

Controller 60

(26) As depicted in FIG. 3, the controller 60 has a CPU 61, a ROM 62, a RAM 63, an EEPROM 64, and an ASIC 65. The ROM 62 stores the likes of various kinds of data required in operation of the controller 60. The RAM 63 is a working memory of the CPU 61. The EEPROM 64 stores the likes of a control program executed by the CPU 61. Prior to image recording being executed by the printer 10, the control program stored in the EEPROM 64 is copied to the RAM 63. The CPU 61 executes the control program stored in the RAM 63. As a result, the controller 60 executes an image recording processing and a wipe processing that will be described later.

(27) The controller 60 is electrically connected to the motor for holder drive 71, the motor for conveyance 72, the motor for carriage drive 73, the motor for cap drive 74, the motor for wiper drive 75, and the head 42, via the ASIC 65. The motor for holder drive 71, the motor for conveyance 72, the motor for carriage drive 73, the motor for cap drive 74, and the motor for wiper drive 75 rotate according to control from the controller 60, and generate motive power. The head 42 discharges ink onto the sheet S conveyed on the platen 25, according to control from the controller 60.

(28) The holder 21 rotates due to motive power from the motor for holder drive 71. The drive rollers 32, 34 rotate due to motive power from the motor for conveyance 72. The sheet S is conveyed in the conveying direction 6 due to motive power from the motor for conveyance 72. The carriage 41 moves in the left-right direction 9 due to motive power from the motor for carriage drive 73. The cap 39 moves in the up-down direction 7 between a relatively high covering position and a relatively low separated position, due to motive power from the motor for cap drive 74. The wiper unit 50 rotates due to motive power from the motor for wiper drive 75, and has its state changed between the three states depicted in FIGS. 5A to 5C. Note that some of the motor for holder drive 71, the motor for conveyance 72, the motor for carriage drive 73, the motor for cap drive 74, and the motor for wiper drive 75 may be realized by a common motor (that is, one motor functioning as two or more of them).

Moving Range of Carriage 41, and Cap 39

(29) As depicted in FIG. 2, the platen 25, which has a shape long in the left-right direction 9, is located below the carriage 41 in the up-down direction 7 (see, FIG. 1). A left end of the platen 25 is located close to left ends of the guide rails 37, 38 in the left-right direction 9. A right end of the platen 25 is located further to the right than centers of the guide rails 37, 38 in the left-right direction 9. The wiper unit 50 is located to the right of the platen 25 in the left-right direction 9. The cap 39 is located further to the right of the wiper unit 50, close to right ends of the guide rails 37, 38 in the left-right direction 9. While image recording is being executed by the printer 10, the carriage 41 moves in the left-right direction 9 within a range of the platen 25. While image recording is not being executed by the printer 10, the carriage 41 is located in a position at which the head 42 will face the cap 39 (hereafter, referred to as a standby position).

(30) When the carriage 41 is located in the standby position, the cap 39 is located in the covering position. The cap 39 located in the covering position covers the nozzle surface 43 of the head 42. When the carriage 41 is located in other than the standby position, the cap 39 is located in the separated position. The cap 39 located in the separated position does not cover the nozzle surface 43 of the head 42. In this way, the cap 39 has a function of covering the nozzle surface 43 of the head 42 while image recording is not being executed.

Wiper Unit 50

(31) As depicted in FIG. 4, the wiper unit 50 has a first wiper 51, a second wiper 52, a wiper supporting portion 53, and a cleaning liquid tub 54. The cleaning liquid tub 54 is shaped like an open-topped box. As depicted in FIGS. 4 and 5A to 5C, the cleaning liquid tub 54, which is located at the lower part of the wiper unit 50, stores a cleaning liquid L. The cleaning liquid L is a liquid suitable for removing an unnecessary object that has adhered to the nozzle surface 43 of the head 42. Glycerin, for example, is used for the cleaning liquid L.

(32) The first wiper **51** is shaped like a rectangular parallelepiped whose length in the front-rear direction **8** is long, and whose length in the left-right direction **9** is short. The length in the front-rear direction **8** of the first wiper **51** is shorter than a length in the front-rear direction **8** of the wiper supporting portion **53**. The second wiper **52** has substantially the same shape and size as the first wiper **51**. However, when cut in a plane orthogonal to the front-rear direction **8**, cross-sectional shape of the first wiper **51** is rectangular, whereas cross-sectional shape of the second wiper **52** is trapezoidal. Hereafter, the first wiper **51** and the second wiper **52** will be collectively referred to as two wipers.

(33) The first wiper **51**, which is capable of being impregnated with the cleaning liquid L, deforms with a high degree of freedom in response to an external force. On the other hand, the second wiper **52**, which is not capable of being impregnated with the cleaning liquid L, deforms while keeping the shape of the second wiper **52** to a certain extent, in response to an external force. The first wiper **51** is formed by a porous material, for example. The second wiper **52** is formed by a rubber material, for example. The first wiper **51** is one example of a wiper. The second wiper **52** is one example of an additional wiper.

(34) The wiper supporting portion **53**, which has a shape whose length in the front-rear direction **8** is long, includes an attachment surface **55** and a rotating shaft **56**. The first wiper **51** is attached to the attachment surface **55** so that its longitudinal direction coincides with the front-rear direction **8**. The second wiper **52** is attached to the attachment surface **55** so as to face the first wiper **51**. The two wipers are attached to the attachment surface **55** so that in a state of the attachment surface **55** facing upward, the first wiper **51** will be located further to the right than the second wiper **52** in the left-right direction **9**.

(35) The cleaning liquid tub **54** has a front wall **57** and rear wall (not depicted) that extend in the up-down direction **7** and the left-right direction **9**. The front wall **57** and rear wall each have a through-hole in their central upper portion (in FIG. 4, only the through-hole **58** of the front wall **57** is depicted). One end of the rotating shaft **56** is inserted in the through-hole **58** of the front wall **57**. The other end (not depicted) of the rotating shaft **56** is inserted in the through-hole of the rear wall. As a result, the wiper supporting portion **53** is supported by the cleaning liquid tub **54** in a manner enabling the wiper supporting portion **53** to rotate around the rotating shaft **56**. Motive power of the motor for wiper drive **75** (see, FIG. 3) is transmitted to a wiper drive mechanism (not depicted). The rotating shaft **56** undergoes a half-rotation due to action of the wiper drive mechanism. As a result, the first wiper **51**, second wiper **52**, and wiper supporting portion **53** also undergo a half-rotation around the rotating shaft **56**. Note that configuration of the wiper drive mechanism may be any configuration.

(36) As depicted in FIGS. 5A to 5C, the wiper unit **50** has its state changed between an upward facing state, an intermediate state, and a downward facing state, due to undergoing a half-rotation around the rotating shaft **56**. The upward facing state (FIG. 5A) is a state of the attachment surface **55** facing upward. The intermediate state (FIG. 5B) is a state of the attachment surface **55** facing rightward. The downward facing state (FIG. 5C) is a state of the attachment surface **55** facing downward.

(37) In the upward facing state, the two wipers are not immersed in the cleaning liquid L stored in the cleaning liquid tub **54**, and are located in a position allowing the two wipers to abut on the nozzle surface **43** of the head **42**. In the downward facing state, part of the first wiper **51** and part of the second wiper **52** are immersed in the liquid L stored in the cleaning liquid tub **54**, and the two wipers are located in a position where the two wipers cannot abut on the nozzle surface **43**. The intermediate state is a state midway between the upward facing state and the downward facing state. In the intermediate state, the two wipers are not immersed in the cleaning liquid L stored in the cleaning liquid tub **54**, and are located in a position where they cannot abut on the nozzle surface **43**. Note that in the downward facing state, the whole of the first wiper **51** and whole of the second wiper **52** may be immersed in the cleaning liquid L.

(38) While image recording is not being performed the carriage **41** is located in the standby position, and the wiper unit **50** is in the downward facing state. In this period, the first wiper **51** is impregnated with the cleaning liquid L stored in the cleaning liquid tub **54**. The controller **60** receives an image recording instruction, and thereupon moves the carriage **41** from the standby position to a recording start position to execute image recording. After receiving the image recording instruction but before executing image recording, the controller **60** uses the wiper unit **50** to execute the wipe processing depicted in FIG. 7.

Operation of Controller **60**

(39) Operation of the controller **60** will be described with reference to FIG. 6. At a timepoint when the controller **60** has reached a step **S11**, the carriage **41** is located in the standby position, the cap **39** is located in the covering position, and the wiper unit **50** is in the downward facing state. At this time, the nozzle surface **43** of the head **42** is covered by the cap **39**, and part of the first wiper **51** and part of the second wiper **52** are immersed in the cleaning liquid L stored in the cleaning liquid tub **54**. The first wiper **51** is in a state of having been impregnated with the cleaning liquid L.

(40) The controller **60** receives an image recording instruction from an operating unit (not depicted) (**S11**). Specifically, the controller **60** waits in the step **S11** until it receives the image recording instruction. Upon receiving the image recording instruction in the step **S11**, the controller **60** moves the cap **39** from the covering position to the separated position (**S12**). Next, the controller **60** executes the wipe processing (details of which will be described later) (**S13**). At a timepoint when the controller **60** has completed processing up to the step **S13**, the carriage **41** is located to the left of the wiper unit **50** in the left-right direction **9**.

(41) Next, the controller **60** moves the carriage **41** in a leftward direction to a recording start position (**S14**). The recording start position is a predetermined position at which the carriage **41** will face the platen **25**. Next, the controller **60** conveys the sheet S to the recording start position (**S15**). Note that the controller **60** may execute the step **S15** parallelly to the whole or part of the steps from the step **S12** to the step **S14**. At a timepoint when the controller **60** has completed processing up to the step **S15**, the printer **10** is in a state allowing image recording to be executed.

(42) Next, the controller **60** executes image recording on the sheet S (**S21**). In the step **S21**, the controller **60** moves the carriage **41** in the left-right direction **9** (in a leftward direction or rightward direction). While the carriage **41** is moving in the left-right direction **9**, the controller **60** causes ink as much as a quantity corresponding to image data to be discharged from the nozzles **44** of the head **42**.

(43) Next, the controller **60** determines whether there is image data remaining (**S22**). In response to having determined there to be image data remaining in the step **S22** (**S22**: Yes), the controller **60** proceeds to a step **S23**. In this case, the controller **60** conveys the sheet S a predetermined amount (**S23**), and proceeds to the step **S21**.

(44) In response to having determined there to be no image data remaining in the step **S22** (**S22**: No), the controller **60** proceeds to a step **S24**. In this case, the controller **60** discharges the sheet S to a predetermined position (**S24**). At a timepoint when the controller **60** has completed processing up to the step **S24**, the printer **10** is in a state of having completed the image recording.

(45) Next, the controller **60** moves the carriage **41** in a rightward direction to the standby position (**S25**). Next, the controller **60** moves the cap **39** in an upward direction from the separated position to the covering position (**S26**). After that, the controller **60** proceeds to the step **S11** in order to execute the next image recording.

(46) Details of the wipe processing executed by the controller **60** will be described with reference to FIGS. 7 and 8A to 8C. At a timepoint when the controller **60** starts execution of the wipe processing, the carriage **41** is located in the standby position, the cap **39** is located in the separated position, and the wiper unit **50** is in the downward facing state.

(47) The standby position is located further to the right than the wiper unit **50** in the left-right direction **9**. At a timepoint when the controller **60** starts execution of the wipe processing, the

carriage **41** is located further to the right than the wiper unit **50** in the left-right direction **9**. As depicted in FIG. **8A**, the nozzle surface **43** of the head **42** includes: a first region **45** not having the nozzles **44**; and a second region **46** having the nozzles **44**.

(48) The controller **60** controls the wiper unit **50** to the upward facing state at the start of the wipe processing (**S31**). Upon the wiper unit **50** changing its state from the downward facing state to the upward facing state, the two wipers emerge (come out) from the cleaning liquid **L** stored in the cleaning liquid tub **54** to be in an upright state (see, FIG. **8A**). In the course of the wiper unit **50** changing its state from the downward facing state to the upward facing state, and while the wiper unit **50** is in the upward facing state, some of the cleaning liquid that the first wiper **51** has been impregnated with, and cleaning liquid that has adhered to a surface of the second wiper **52** moves in a downward direction due to action of gravity, and is collected in the cleaning liquid tub **54**.

(49) Next, the controller **60** moves the carriage **41** in a leftward direction to a pressing-against position (**S32**). The pressing-against position is a position of the carriage **41** by which the first wiper **51** will abut on the first region **45** (the region not having the nozzles **44**) of the nozzle surface **43** (see, FIG. **8B**). The pressing-against position may be a position of the carriage **41** at which, for example, the first wiper **51** will abut on the first region **45** and the second wiper **52** will abut on a side surface **47** of the head **42**. By setting the pressing-against position to the above-described position, the cleaning liquid that has adhered to the head **42** when the head **42** is abutted on by the first wiper **51** can be recovered in the cleaning liquid tub **54** via the second wiper **52**. Note that, a moving speed of the carriage **41** until it reaches the pressing-against position may be any speed.

(50) Next, the controller **60** keeps the carriage **41** in a stopped state for a first time period at the pressing-against position (**S33**). The step **S33** is one example of pressing-against processing. While the carriage **41** is stopped at the pressing-against position, the first wiper **51** abuts on the first region **45**, and the first wiper **51** deforms with its tip portion bending to the left (see, FIG. **8B**). As a result, the first wiper **51** is squeezed, and some of the cleaning liquid with which the first wiper **51** has been impregnated separates from the first wiper **51**. The cleaning liquid that has separated from the first wiper **51** moves in a downward direction due to action of gravity, and is recovered in the cleaning liquid tub **54**. Length of the first time period is determined according to the likes of material (quality of material) of the first wiper **51**, characteristics of the cleaning liquid, and degree of deformation of the first wiper **51**, for example. The first time period is one second or more, for example.

(51) Next, the controller **60** moves the carriage **41** in a leftward direction at a speed suitable for wiping the nozzle surface **43** of the head **42** (hereafter, referred to as a wipe speed), for a second time period (**S34**). The step **S34** is one example of a moving processing. While the step **S34** is being executed by the controller **60**, the first wiper **51** abuts on the second region **46** (the region having the nozzles **44**) of the nozzle surface **43**. At this time, the two wipers abut on the second region **46**, and the two wipers deform with their tip portions bending to the left (see, FIG. **8C**). Thus, due to the carriage **41** moving in a leftward direction in a state of the second region **46** having been abutted on by the two wipers, an unnecessary object that has adhered to the nozzle surface **43** of the head **42** is removed.

(52) Note that in the step **S34**, the controller **60** accelerates the carriage **41** in the stopped-state to the wipe speed, and then moves the carriage **41** at the wipe speed for the second time period. During the second time period, a moving speed of the carriage **41** is constant (is always the wipe speed), and a time period that the carriage **41** is accelerating is not included in the second time period.

(53) Next, the controller **60** controls the wiper unit **50** to the downward facing state (**S35**). Upon the wiper unit **50** changing its state from the upward facing state to the downward facing state, part of the first wiper **51** and part of the second wiper **52** are again immersed in the cleaning liquid **L** stored in the cleaning liquid tub **54** (see, FIG. **5C**). Subsequently, the controller **60** executes the processing of the step **S14** and the steps following the step **S14** depicted in FIG. **6**.

Effects of Embodiment

(54) As described above, the controller **60** executes: the pressing-against processing in which the carriage **41** is kept in a stopped state at the pressing position for the first time period, and the first wiper **51** is abutted on the first region **45**; and the moving processing in which, while the first wiper **51** is abutting on the second region **46**, the carriage **41** is moved in a leftward direction at the wipe speed.

(55) The printer **10** and controller **60** according to the present embodiment result in that by the pressing-against processing being executed, the first wiper **51** that has been impregnated with the cleaning liquid is squeezed, and the cleaning liquid with which the first wiper **51** has been impregnated is reduced to a suitable amount. The pressing-against processing can be executed without a dedicated member being employed. Hence, by the moving processing being executed subsequently to the pressing-against processing, wiper processing employing a suitable amount of cleaning liquid can be executed by a simple configuration.

(56) When relative moving speed of the carriage **41** and first wiper **51** during pressing-against processing is assumed to be the first speed, and relative moving speed of the carriage **41** and first wiper **51** during moving processing is assumed to be the second speed, the first speed is zero, and the second speed is the wipe speed. The second speed is faster than the first speed. Since the first speed is zero, the first wiper **51** is squeezed in a state of the carriage **41** and first wiper **51** having been relatively stopped.

(57) In the moving processing, the controller **60** moves the carriage **41** in a leftward direction along the left-right direction **9** at the wipe speed, in a state of the second wiper **52** having been abutted on the second region **46**. This results in that after the nozzle surface **43** has been applied with a suitable amount of the cleaning liquid using the first wiper **51**, the cleaning liquid and an unnecessary object that have adhered to the nozzle surface **43** can be removed using the second wiper **52**.

(58) By the controller **60** executing the pressing-against processing in a state of the second wiper **52** having been abutted on the side surface **47** of the head **42**, the cleaning liquid that has adhered to the head **42** can be removed via the second wiper **52**. Due to the first time period being set at one second or more, the first wiper **51** is squeezed for a sufficient time.

(59) While the invention has been described in conjunction with various example structures outlined above and illustrated in the figures, various alternatives, modifications, variations, improvements, and/or substantial equivalents, whether known or that may be presently unforeseen, may become apparent to those having at least ordinary skill in the art. Accordingly, the example embodiments of the disclosure, as set forth above, are intended to be illustrative of the invention, and not limiting the invention. Various changes may be made without departing from the spirit and scope of the disclosure. Therefore, the disclosure is intended to embrace all known or later developed alternatives, modifications, variations, improvements, and/or substantial equivalents. Some specific examples of potential alternatives, modifications, or variations in the described invention are provided below:

Modifications

(60) In the above-described embodiment, the controller **60** keeps the carriage **41** in a stopped state for the first time period in the pressing-against processing. In a first modification, the controller **60** moves the carriage **41** at a speed greater than zero for the first time period in the pressing-against processing. That is, in the first modification, the first speed (the relative moving speed of the carriage **41** and first wiper **51** during the pressing-against processing) is greater than zero. In such a first modification too, the first wiper **51** that has been impregnated with the cleaning liquid is squeezed, and the cleaning liquid with which the first wiper **51** has been impregnated is reduced to a suitable amount, in the pressing-against processing. Since the first speed is greater than zero, the first wiper **51** is squeezed in a state of the carriage **41** and first wiper **51** moving relative to each other.

(61) In the above-described embodiment, controller **60** moves the carriage **41** in the left-right direction **9** (specifically, in a leftward direction), without moving the two wipers in the left-right direction **9**, when executing wipe processing. In a second modification, the controller **60** moves the two wipers in the left-right direction **9**, without moving the carriage **41** in the left-right direction **9**, when executing wipe processing. Alternatively, the controller **60** moves both the carriage **41** and two wipers in the left-right direction **9**, when executing wipe processing.

(62) Generally, in the pressing-against processing, the controller **60** may move the carriage **41** and first wiper **51** relative to each other in the left-right direction **9** for the first time period at the first speed, while abutting the first wiper **51** on the first region **45**. In the moving processing, the controller **60** may move the carriage **41** and first wiper **51** relative to each other in the left-right direction **9** at the second speed faster than the first speed, while abutting the first wiper **51** on the second region **46**.

(63) The controller **60** may move the carriage **41** and first wiper **51** relative to each other in a moving direction in the pressing-against processing, and then, move the carriage **41** and first wiper **51** relative to each other in an opposite direction opposite to the moving direction of the pressing-against processing in the moving processing. Meanwhile the controller **60** may move the carriage **41** and first wiper **51** relative to each other in the moving processing without moving the carriage **41** and first wiper **51** relative to each other in the opposite direction. By doing so, the moving processing can be executed immediately after the pressing-against processing.

(64) In the above-described embodiment, controller **60** executes the wipe processing depicted in FIG. **7**, in the step **S13**. In a third modification, the controller **60** executes the wipe processing depicted in FIG. **9**, in the step **S13**. The wipe processing depicted in FIG. **9** is the wipe processing depicted in FIG. **7** to which step **S41-S44** have been added.

(65) In the wipe processing depicted in FIG. **9**, the controller **60** execute the steps **S31-S34**, after which the controller **60** controls the wiper unit **50** to the intermediate state (**S41**). In the intermediate state, the two wipers are not immersed in the cleaning liquid **L** stored in the cleaning liquid tub **54**, and are located in a position where the two wipers are unable to abut on the nozzle surface **43** of the head **42**.

(66) Next, the controller **60** moves the carriage **41** in a rightward direction to the pressing-against position (**S42**). The pressing-against position is a position of the carriage **41** when moving processing is started. Note that in the step **S42**, the controller **60** may move the carriage **41** to the position where moving processing is started or to a position separated rightward from the position where moving processing is started.

(67) Next, the controller **60** controls the wiper unit **50** to the upward facing state (**S43**). Next, the controller **60** moves the carriage **41** in a leftward direction at the wipe speed for the second time period (**S44**). Processing of the step **S44** is the same as processing of the step **S34**. The controller **60** does not execute pressing-against processing in the process between the step **S42** and the step **S44**. Next, the controller **60** controls the wiper unit **50** to the downward facing state (**S35**).

(68) In the third modification, subsequent to moving processing, the controller **60** moves the carriage **41** to the position where moving processing is started or to a position further from the position where moving processing is ended as compared with the position where moving processing is started, without causing the two wipers to abut on the head **42**, and re-executes the moving processing. In the third modification, pressing-against processing is executed prior to executing moving processing for the first time, so an amount of the cleaning liquid with which the first wiper **51** is impregnated will be still suitable when executing moving processing for the second time. Hence, the nozzle surface **43** of the head **42** can be wiped twice using a suitable amount of the cleaning liquid.

(69) In the above-described embodiment, the printer **10** includes the wiper unit **50** depicted in FIG. **4**. In a fourth modification, the printer **10** includes a wiper unit **59** depicted in FIG. **10**. The wiper unit **59** depicted in FIG. **10** has the first wiper **51**, the wiper supporting portion **53**, and the cleaning

liquid tub **54**, but does not have the second wiper **52**. Thus, the printer need not include a second wiper not capable of being impregnated with the cleaning liquid.

(70) In the above-described embodiment, in the moving processing after the pressing-against processing, the controller **60** moves the carriage **41** in the same direction (leftward direction) as a moving direction of the carriage **41** prior to pressing-against processing, without moving the carriage **41** in an direction (rightward direction) opposite to the moving direction of the carriage **41** prior to pressing-against processing. In a fifth modification, the controller **60** moves the carriage **41** in the direction (rightward direction) opposite to the moving direction of the carriage **41** prior to pressing-against processing for a predetermined time, after which the controller **60** moves the carriage **41** in the same direction (leftward direction) as the moving direction of the carriage **41** prior to pressing-against processing, in the moving processing after the pressing-against processing.

(71) In the above-described embodiment, the printer **10** includes one head **42**. However, the printer **10** may include two or more heads **42**. In the above-described embodiment, the printer **10** includes four tanks **26A-26D**. However, the printer **10** may include three or fewer, or five or more tanks.

(72) Moreover, characteristics of the printers and controllers according to the above-described embodiment and its modifications may be arbitrarily combined provided there occurs no contradiction with those characteristics, and there may thereby be configured printers and controllers that combine the characteristics of the printers and controllers according to the above-described embodiments and modifications.

Claims

1. A controller for an image recording device, the image recording device including: a carriage capable of moving in a first direction; a head which is mounted on the carriage and which is configured to discharge a liquid from a nozzle; and a wiper capable of being impregnated with a cleaning liquid, a nozzle surface of the head including a first region not having the nozzle and a second region having the nozzle, the controller being configured to execute: a pressing-against processing of abutting the wiper on the first region in a state in which a speed of a relative movement in the first direction between the carriage and the wiper is a first speed; and then a moving processing of moving the carriage and the wiper relative to each other in the first direction at a second speed faster than the first speed, while abutting the wiper on the second region, after the executing of the pressing-against processing, contact between the wiper and the head being maintained from an end of the pressing-against processing to a beginning of the moving processing.
2. The controller according to claim 1, wherein the first speed is zero.
3. The controller according to claim 1, wherein the first speed is greater than zero.
4. The controller according to claim 1, wherein the image recording device further includes an additional wiper not capable of being impregnated with the cleaning liquid; and in the moving processing, the carriage and the additional wiper are moved relative to each other in the first direction at the second speed, in a state in which the additional wiper abuts on the second region.
5. The controller according to claim 4, wherein the pressing-against processing is executed in a state in which the additional wiper abuts on a side surface of the head.
6. The controller according to claim 1, wherein: in the pressing-against processing, the carriage and the wiper are moved relative to each other in a moving direction; and then, in the moving processing, the carriage and the wiper are moved relative to each other in the moving direction, without being moved relative to each other in a direction opposite to the moving direction.
7. The controller according to claim 1, wherein the controller is configured to: after the moving processing, move the carriage and the wiper relative to each other, in a state in which the wiper does not abut on the head, until the wiper is located relative to the head at a starting position or at a position further from an ending position as compared with the starting position, the starting

position being a position of the wiper relative to the head when the moving processing is started, the ending position being a position of the wiper relative to the head when the moving processing is ended; and execute the moving processing again.

8. The controller according to claim 1, wherein the controller is configured to execute the pressing-against processing for one second or more.

9. A control method for an image recording device, the image recording device including: a carriage capable of moving in a first direction; a head which is mounted on the carriage and which is configured to discharge a liquid from a nozzle; and a wiper capable of being impregnated with a cleaning liquid, a nozzle surface of the head including a first region not having the nozzle and a second region having the nozzle, the control method comprising: executing a pressing-against processing of abutting the wiper on the first region in a state in which a speed of a relative movement in the first direction between the carriage and the wiper is a first speed; and then executing a moving processing of moving the carriage and the wiper relative to each other in the first direction at a second speed faster than the first speed, while abutting the wiper on the second region, after the executing of the pressing-against processing, contact between the wiper and the head being maintained from an end of the pressing-against processing to a beginning of the moving processing.

10. An image recording device comprising: a carriage capable of moving in a first direction; a head which is mounted on the carriage and which is configured to discharge a liquid from a nozzle; a wiper capable of being impregnated with a cleaning liquid; and a controller as defined in claim 1, wherein a nozzle surface of the head includes a first region not having a nozzle and a second region having a nozzle.
